10TH CLASS CHEMISTRY — NEW SYLLABUS 2025-26 (PECTAA)

Chapter 16 — Electrochemistry

Solved Quick Checks + Exercise | New Book (PECTAA 2025-26)

Quick Checks (Solved)

Quick Check 16.1

1. Assign oxidation number to the marked atom in the following compounds: $K_2Cr_2O_7$ (Cr) and $(NH_4)_2SO_4$ (S).

(i) $K_2Cr_2O_7$: potassium is +1 and oxygen is −2. The sum of the oxidation numbers in a neutral compound is zero.

$2(+1) + 2Cr + 7(-2) = 0 \rightarrow 2 + 2Cr - 14 = 0 \rightarrow 2Cr = +12 \rightarrow$ **Cr = +6**

(ii) $(NH_4)_2SO_4$: the ammonium ion is NH_4^+, so nitrogen in it is −3 and hydrogen is +1; oxygen is −2.

$2(-3) + 8(+1) + S + 4(-2) = 0 \rightarrow -6 + 8 + S - 8 = 0 \rightarrow$ **S = +6**

So chromium in $K_2Cr_2O_7$ has oxidation number +6 and sulphur in ammonium sulphate has oxidation number +6.

Quick Check 16.2

1. Write the formulae of the following ionic compounds with the help of oxidation numbers of their constituent atoms: aluminium nitride, sodium sulphite and sodium hydrogen carbonate.

Aluminium nitride: aluminium is in group 13, so its oxidation number is +3; nitrogen is more electronegative and takes −3. The formula that makes the sum zero is **AlN**, because (+3) + (−3) = 0.

Sodium sulphite: sodium is +1 and the sulphite ion is SO_3^{2-}, so its oxidation number is −2. Two sodium ions are needed for one sulphite ion, so the formula is **Na_2SO_3**.

Sodium hydrogen carbonate: sodium is +1 and the hydrogen carbonate ion is HCO_3^-, so its oxidation number is −1. One sodium ion balances one hydrogen carbonate ion, so the formula is **$NaHCO_3$**.

2. Write the formulae of the following ionic compounds with the help of the charges present on their ions: ammonium acetate, calcium phosphate and potassium chromate.

Ammonium acetate: NH_4^+ and CH_3COO^- carry equal but opposite charges, so the formula is **NH_4CH_3COO** (or CH_3COONH_4).

Calcium phosphate: Ca^{2+} and PO_4^{3-}; three calcium ions give +6 and two phosphate ions give −6, so the formula is **$Ca_3(PO_4)_2$**.

Potassium chromate: K^+ and CrO_4^{2-}; two potassium ions balance one chromate ion, so the formula is **K_2CrO_4**.

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Quick Check 16.3

1. Identify the oxidising agent and the reducing agent in the following reactions: (i) $2CuSO_4 + 4KI \rightarrow 2CuI + I_2 + 2K_2SO_4$ (ii) $H_2S + Cl_2 \rightarrow S + 2HCl$

(i) Copper in $CuSO_4$ goes from +2 to +1, so it gains electrons and is reduced; therefore $CuSO_4$ is the **oxidising agent**. Iodine in KI goes from −1 to 0, so it loses electrons and is oxidised; therefore KI is the **reducing agent**.

(ii) Sulphur in H_2S goes from −2 to 0, so it loses electrons and is oxidised; therefore H_2S is the **reducing agent**. Chlorine goes from 0 to −1, so it gains electrons and is reduced; therefore Cl_2 is the **oxidising agent**.

Quick Check 16.4

1. Why are carbon electrodes used in the electrolysis of concentrated solution of NaCl?

Carbon (graphite) is an inert electrode: it conducts electricity but it does not take part in the reaction, and it is cheap and easily available. In the electrolysis of concentrated brine, chlorine gas is produced at the anode, and chlorine is very corrosive. Graphite does not react with the chlorine produced, while many metal anodes would be attacked and destroyed by it. That is why the anode of this cell is made of graphite, while the cathode is made of platinum.

Quick Check 16.5

1. Which other electrolyte can be used in a hydrogen-oxygen fuel cell?

Aqueous sodium hydroxide (NaOH) solution can be used in place of aqueous potassium hydroxide (KOH). Any strong alkali can act as the electrolyte because it gives freely moving hydroxide ions (OH^-), which carry the charge between the two electrodes. These ions are used up at the anode and produced again at the cathode, so the electrolyte keeps working.

Quick Check 16.6

1. What conditions are required for rusting of iron?

Both oxygen (air) and water (moisture) must be present at the surface of iron. In dry air without moisture, or in boiled water where air has been removed, iron does not rust. Rusting also becomes faster when the air is damp, when salt is present, and when the temperature is high, which is why iron objects rust quickly in the rainy season and in sea-coast areas.

2. If a sample of iron and a sample of zinc come into contact, which will corrode? Explain your answer.

Zinc will corrode and the iron will remain protected. Zinc is more reactive than iron and lies above it in the electrochemical series, so zinc loses electrons more easily and is oxidised in place of iron. The electrons lost by zinc flow to the iron and stop iron from losing its own electrons. This method of protection is called **sacrificial protection**, and it is used in galvanised iron sheets and in ships where blocks of zinc or magnesium are fixed to the steel hull.

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Quick Check 16.7

1. What is the principle behind electroplating?

Electroplating works on the principle of electrolysis. The article to be plated is cleaned and made the cathode (negative electrode), and a plate of the plating metal is made the anode (positive electrode). The electrolyte is a salt solution of the plating metal. When current is passed, the metal ions of the electrolyte move to the cathode, gain electrons and deposit as a thin, shiny layer on the article. At the same time the anode slowly dissolves and keeps the concentration of the electrolyte constant.

2. How is electroplating used to repair worn-out areas of metal objects?

A worn-out or broken part is made the cathode in a suitable electrolyte, and the metal that is to be filled in is made the anode. On passing current, metal ions deposit on the worn part and fill the gaps, so the missing metal is built up again. This is called electrodeposition or electro-repair. It is used to repair worn parts of machinery, printing plates and decorative articles, and the method saves the cost of replacing the whole part.

Quick Check 16.8

1. Compare an electrolytic cell and a galvanic cell.

Electrolytic cell	Galvanic (voltaic) cell
It converts electrical energy into chemical energy.	It converts chemical energy into electrical energy.
An external battery is needed to drive the non-spontaneous reaction.	No external battery is needed; the reaction is spontaneous.
The anode is positive and the cathode is negative.	The anode is negative and the cathode is positive.
Oxidation still occurs at the anode and reduction at the cathode.	Oxidation occurs at the anode and reduction at the cathode.
Example: electrolysis of molten lead(II) chloride, electroplating.	Example: Daniel cell (Zn–Cu cell), which gives about 1.10 V.

2. Write down the reactions which occur in a Zn–Mg galvanic cell.

Magnesium is more reactive than zinc and lies above it in the electrochemical series, so magnesium is oxidised at the anode and zinc ions are reduced at the cathode.

At anode (oxidation): $Mg(s) \rightarrow Mg^{2+}(aq) + 2e^{-}$

At cathode (reduction): $Zn^{2+}(aq) + 2e^{-} \rightarrow Zn(s)$

Overall reaction: $Mg(s) + Zn^{2+}(aq) \rightarrow Mg^{2+}(aq) + Zn(s)$

The electrons flow from the magnesium electrode to the zinc electrode through the external wire, and this flow of electrons is the electric current.

Exercise (Solved)

A. Multiple Choice Questions

1. The oxidation number of Mn in K_2MnO_4:

(a) +7 (b) +5

(c) +6 (d) −6

Answer: (c) — 2(+1) + Mn + 4(−2) = 0, so Mn − 6 = 0 and Mn = +6.

2. Which element in the following reaction is being reduced? $4HBr_{(aq)} + MnO_{2(s)} \rightarrow MnBr_{2(aq)} + Br_2 + 2H_2O_{(l)}$

(a) Br (b) Mn

(c) H (d) Both Mn and H

Answer: (b) — Manganese goes from +4 in MnO_2 to +2 in $MnBr_2$, so it gains electrons and is reduced.

3. During the electrolysis of dilute H_2SO_4 usingPt electrodes, which process takes place at the cathode?

(a) Reduction (b) Neither oxidation nor reduction

(c) Oxidation (d) First oxidation and then reduction

Answer: (a) — The cathode is the negative electrode where ions gain electrons, so hydrogen ions are reduced to hydrogen gas.

4. Which product will you get at the anode when an aqueous solution of $CuSO_4$ is electrolysed using copper electrodes?

(a) Cu (b) O_2

(c) H_2 (d) No product; instead copper metal will go into the solution as Cu^{2+}.

Answer: (d) — With copper electrodes the copper anode itself is oxidised and dissolves as Cu^{2+}; no gas is produced at the anode.

5. Which of the following statements is NOT correct about a Zn–Cu electrochemical cell?

(a) Anode is negatively charged (b) Reduction occurs at anode

(c) Cathode is positively charged (d) Reduction occurs at cathode

Answer: (b) — Oxidation always takes place at the anode; reduction takes place at the cathode.

6. Which product will be obtained at the anode when a concentrated solution of NaCl is electrolysed?

(a) Cl_2 (b) O_2

(c) More O_2 and less Cl_2 (d) More Cl_2 and less O_2

Answer: (a) — In concentrated brine the chloride ions are in high concentration, so Cl^- is preferentially discharged and chlorine gas is obtained at the anode.

7. When a dilute aqueous solution of $ZnCl_2$ is electrolysed, which product will you get at the cathode?

(a) H_2 (b) Zn

(c) O_2 (d) Zn and O_2

Answer: (a) — Zinc is above hydrogen in the electrochemical series, so hydrogen ions are discharged more easily and hydrogen gas is obtained at the cathode.

8. What is produced at the cathode in a hydrogen–oxygen fuel cell?

(a) H^+ (b) O^{2-}

(c) H_2O (d) OH^-

Answer: (d) — At the cathode oxygen gains electrons in the presence of water and forms hydroxide ions: $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$.

9. Which of the following metals is highest in the electrochemical series?

(a) Tin (b) Iron

(c) Magnesium (d) Zinc

Answer: (c) — Magnesium has the most negative standard potential (−2.37 V) of the four metals, so it lies highest in the series.

B. Short Answer Questions

16.1 What happens when electricity is passed through an aqueous solution of NaCl?

The solution is electrolysed and decomposes into its constituents. Sodium chloride gives Na^+ and Cl^- ions and water gives a small number of H^+ and OH^- ions. At the cathode hydrogen ions are reduced to hydrogen gas, because hydrogen is discharged more easily than sodium. At the anode, chlorine gas is obtained from concentrated brine, while oxygen is obtained from dilute solution. Sodium hydroxide is left in the solution, so the solution gradually becomes alkaline.

$$2NaCl(aq) + 2H_2O(l) \rightarrow H_{2(g)} + Cl_{2(g)} + 2NaOH(aq)$$

16.2 What are the main objectives of electroplating?

There are three main objectives of electroplating.

(i) Decoration: to deposit noble metals like gold and silver on cheap and ordinary metals, so that the articles look beautiful and expensive.

(ii) Protection: to protect the metals from rusting and from the attack of organic acids, for example chromium plating on iron and tin coating on steel cans.

(iii) Repair: to weld or rebuild the broken and worn-out parts of machinery by electrodepositing metal on them.

16.3 Mention one difference between an electrolytic cell and a galvanic cell.

An electrolytic cell uses electrical energy from an outside source to carry out a non-spontaneous reaction, whereas a galvanic cell uses a spontaneous redox reaction to produce electrical energy. That is why an electrolytic cell needs a battery, while a galvanic cell works by itself and can light a bulb or run a small device.

16.4 Through which electrode do electrons leave the electrolytic cell?

Electrons enter the cell through the anode and leave it through the cathode, which is the negative electrode. At the anode the negative ions lose electrons and these electrons travel through the external wire; at the cathode the positive ions gain electrons from the electrode. Inside the electrolyte, however, the current is carried by the movement of ions and not by electrons.

16.5 How does an electrolyte conduct electricity?

An electrolyte conducts electricity through the movement of its freely moving ions. When a voltage is applied, the positive ions (cations) move towards the cathode and the negative ions (anions) move towards the anode. At the electrodes these ions lose or gain electrons and are discharged as neutral atoms or molecules. Electrons do not travel through the solution; they flow only in the external circuit.

16.6 What will happen if a strip of Cu metal is dipped in $FeSO_4$ solution?

No reaction will take place. Copper lies below iron in the electrochemical series, so copper is less reactive than iron and cannot displace iron from its salt solution. Copper atoms cannot lose electrons to Fe^{2+} ions, so the blue colour of the solution and the mass of the copper strip remain unchanged. If the experiment is reversed and an iron strip is dipped in copper sulphate solution, then iron displaces copper and the blue colour fades.

16.7 Why are fuel cells regarded as environment friendly?

The only chemical product of a hydrogen–oxygen fuel cell is water, so it gives zero emission of carbon dioxide, nitrogen oxides and sulphur oxides, which are produced by petrol and diesel engines and are harmful to the environment. Fuel cells also convert a large part of the chemical energy of the fuel into electrical energy, so they are more efficient, and they produce much less noise pollution. The hydrogen used in them can be obtained from renewable sources such as solar and wind power, which makes them sustainable.

C. Constructed Response Questions

16.1 Calculate the oxidation number of chlorine atom in $Ca(ClO_3)_2$.

Calcium is in group 2, so its oxidation number is +2, and oxygen is taken as −2. The sum of the oxidation numbers of a neutral compound is zero, and there are two chlorine atoms in the formula.

$(+2) + 2Cl + 6(-2) = 0 \rightarrow 2 + 2Cl - 12 = 0 \rightarrow 2Cl = +10$

Cl = +5, so the oxidation number of each chlorine atom in calcium chlorate is +5.

16.2 How can H_2O_2 behave as an oxidizing agent?

In hydrogen peroxide the oxidation number of oxygen is −1, which lies between 0 and −2, so it can either lose electrons (be oxidised to O_2) or gain electrons (be reduced to water). When it acts as an oxidising agent, its oxygen gains electrons and its oxidation number falls from −1 to −2, and in doing so it oxidises the other substance.

$$H_2O_2(aq) + 2H^+(aq) + 2e^- \rightarrow 2H_2O(l) \quad \text{(oxygen } -1 \rightarrow -2\text{, reduction)}$$

For example, it oxidises hydrogen sulphide to sulphur and lead(II) sulphide to lead(II) sulphate. H_2O_2 is a strong oxidising agent because of its high standard reduction potential of +1.77 V.

16.3 Why are different products formed at the electrodes when dilute and concentrated aqueous solutions of NaCl are electrolysed?

In an aqueous solution of sodium chloride both Na^+, Cl^- and the H^+, OH^- ions of water are present, so there is a competition at each electrode, and the ion discharged depends on its concentration and on its position in the electrochemical series.

At the **cathode**, hydrogen is discharged from both dilute and concentrated solutions, because hydrogen is discharged more easily than sodium (sodium lies above hydrogen in the series). So hydrogen gas is obtained at the cathode in both cases.

At the **anode**, the product depends on concentration. In dilute solution, the concentration of Cl^- is very small while OH^- ions are relatively more, so OH^- is preferentially discharged and **oxygen** gas is produced. In concentrated brine, Cl^- ions are present in large amount, so they are preferentially discharged and **chlorine** gas is produced at the anode.

16.4 Why corrosion becomes fast during rainy season?

Rusting of iron needs both oxygen and water. During the rainy season the air is full of moisture and the iron surfaces remain wet for a long time, so a thin film of water containing dissolved oxygen stays on the metal. This film acts as an electrolyte and the electrochemical reactions of rusting take place quickly. Rust is porous, so air and moisture reach the fresh metal under it and the process continues. That is why iron objects corrode much faster in the rainy season than in dry weather.

16.5 If a galvanic cell is established between Cu and Ag, what reactions do you expect at the electrodes?

Copper lies above silver in the electrochemical series, so copper is more reactive, loses electrons and is oxidised at the anode, while silver ions gain electrons and are reduced at the cathode.

At anode: $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$ (oxidation)

At cathode: $2Ag^+(aq) + 2e^- \rightarrow 2Ag(s)$ (reduction)

Overall: $Cu(s) + 2Ag^+(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$

Silver is deposited on the cathode and the copper anode slowly dissolves, so the mass of the anode decreases and that of the cathode increases.

16.6 What is the function of a porous partition in a galvanic cell?

The porous partition keeps the two half-cell solutions separate so that they do not mix directly, but its pores allow ions to pass through it. In this way it completes the internal circuit and maintains the electrical neutrality of the two solutions: the negative ions move towards the anode compartment and the positive ions towards the cathode compartment. Without the partition the two solutions would mix and the cell would not work properly.

16.7 How do fuel cells produce electricity?

A fuel cell has two hollow porous carbon electrodes saturated with platinum, which acts as a catalyst, dipped in aqueous potassium hydroxide. Hydrogen gas is supplied at the anode and oxygen at the cathode.

At anode: $H_{2(g)} + 2OH^{-}(aq) \rightarrow 2H_2O(l) + 2e^{-}$

At cathode: $O_{2(g)} + 2H_2O(l) + 4e^{-} \rightarrow 4OH^{-}(aq)$

Overall: $2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O(l)$

Hydrogen is oxidised at the anode and gives up electrons; these electrons travel through the external circuit to the cathode and this flow is the electric current. The cell produces electricity and water continuously as long as the gases are supplied, and it converts about 75 % of the bond energy of the fuel into electricity.

16.8 How can the chemical reactivity of metals be compared by using the electrochemical series?

The reactivity of a metal depends on how readily it loses electrons to form positive ions. In the electrochemical series the metals are arranged in order of their standard electrode potentials: metals placed higher in the series have more negative potentials and lose electrons more easily, so they are more reactive, while metals placed lower have positive potentials, lose electrons with difficulty and are less reactive.

When a galvanic cell is made between two metals, the more reactive metal is oxidised and becomes the anode, and the less reactive metal is deposited and becomes the cathode. The voltage of the cell also shows the difference in their tendencies to lose electrons: a cell of magnesium and copper gives a higher voltage than a cell of iron and copper, which shows that magnesium is more reactive than iron. By measuring the voltages of different pairs of metals, the whole reactivity series can be arranged.

D. Descriptive Questions

16.1 Explain with examples how the formula of an ionic compound can be written using oxidation numbers.

The formula of an ionic compound is written so that the sum of the oxidation numbers of all the atoms in it becomes zero, because a compound is electrically neutral.

Example 1 – calcium chloride: calcium is in group 2, so its oxidation number is +2, and chlorine, being a group 17 element combined with a less electronegative element, has oxidation number −1. One Ca^{2+} needs two Cl^{-} to make the total zero, so the formula is **$CaCl_2$**.

Example 2 – aluminium oxide: aluminium is in group 13, so its oxidation number is +3, and oxygen is −2. Two aluminium atoms give +6 and three oxygen atoms give −6, so the formula is **Al_2O_3**.

Example 3 – aluminium sulphate: aluminium is +3 and the sulphate ion is SO_4^{2-}; two Al^{3+} give +6 and three SO_4^{2-} give −6, so the formula is **$Al_2(SO_4)_3$**. When more than one polyatomic ion is needed, its formula is written in brackets with a subscript outside.

16.2 Find the oxidation numbers of the indicated elements in the following compounds: $Al(NO_3)_3$ (N), $H_2S_2O_7$ (S) and $Na_2S_2O_3$ (S).

$Al(NO_3)_3$: aluminium is +3 and each nitrate ion carries −1, so for nitrogen in NO_3^-: N + 3(−2) = −1, that is N − 6 = −1, so **N = +5**.

$H_2S_2O_7$: 2(+1) + 2S + 7(−2) = 0, so 2S − 12 = 0 and **S = +6**.

$Na_2S_2O_3$: 2(+1) + 2S + 3(−2) = 0, so 2S − 4 = 0 and **S = +2**. This is the average oxidation number of the two sulphur atoms in thiosulphate.

16.3 Write down the formulae of the following compounds using ionic charges: sodium hypochlorite, calcium oxalate, ammonium phosphate and calcium nitrite.

Sodium hypochlorite: Na^+ and ClO^- give **NaClO**.

Calcium oxalate: Ca^{2+} and $C_2O_4^{2-}$ give **CaC_2O_4**.

Ammonium phosphate: NH_4^+ and PO_4^{3-}; three ammonium ions are needed for one phosphate ion, so the formula is **$(NH_4)_3PO_4$**.

Calcium nitrite: Ca^{2+} and NO_2^- need two nitrite ions for one calcium ion, so the formula is **$Ca(NO_2)_2$**.

16.4 Describe the working of a fuel cell.

A hydrogen–oxygen fuel cell consists of two hollow porous carbon electrodes saturated with platinum, which acts as a catalyst. Both electrodes dip in an electrolyte, which is aqueous potassium hydroxide. Hydrogen gas is supplied to the anode and oxygen gas to the cathode.

At the anode hydrogen reacts with hydroxide ions and is oxidised to water, giving up electrons. These electrons flow through the external circuit to the cathode, and this flow of electrons is the electric current that runs the load. At the cathode oxygen takes these electrons and is reduced to hydroxide ions, which then move back to the anode. Water is formed as the only chemical product.

$$\text{At anode:}\quad 2H_{2(g)} + 4OH^-(aq) \rightarrow 4H_2O(l) + 4e^-$$

$$\text{At cathode:}\quad O_{2(g)} + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$$

$$\text{Overall:}\quad 2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O(l)$$

The cell produces electricity and water continuously as long as the reactants are supplied. It is very efficient, because it converts about 75 % of the bond energy of the fuel into electricity.

16.5 Discuss the electrolysis of concentrated solutions of metal halides.

In a concentrated aqueous solution of a metal halide, the ions of the metal and of the halide are present along with a small number of H^+ and OH^- ions of water. The ion that is discharged at each electrode depends on its position in the electrochemical series and on its concentration.

At the cathode: if the metal lies above hydrogen in the electrochemical series (Na, Mg, Al), its ions are not discharged; instead hydrogen ions are reduced and hydrogen gas is produced. If the metal lies below hydrogen (Cu, Ag), the metal ions are discharged and the metal is deposited on the cathode.

At the anode: the halide ions are present in high concentration, so they are preferentially discharged in place of OH^- ions and halogen gas is produced: $2X^- \rightarrow X_2 + 2e^-$.

Example – concentrated NaCl (brine): hydrogen is obtained at the cathode, chlorine at the anode and sodium hydroxide is left in the solution.

$$2NaCl(aq) + 2H_2O(l) \rightarrow H_{2(g)} + Cl_{2(g)} + 2NaOH(aq)$$

Example – concentrated $CuCl_2$: copper is deposited at the cathode and chlorine gas is produced at the anode, because copper lies below hydrogen in the series.

16.6 Explain electroplating giving examples.

Electroplating is the process in which a thin layer of one metal is deposited on another metal by electrolysis. Its three main objectives are decoration, protection from rusting and repair of worn-out parts.

Procedure: the article to be plated is first cleaned with sand, then washed with caustic soda solution and finally with plenty of water. It is made the cathode, and a plate of the coating metal is made the anode. The electrolyte is a salt solution of the coating metal, and the process is carried out in an electrolytic tank. On passing current, metal ions move to the cathode, gain electrons and deposit as a thin layer, while the anode slowly dissolves and keeps the electrolyte concentration constant.

Silver plating: the article is made the cathode, a silver plate is the anode and potassium dicyanoargentate(I), $K[Ag(CN)_2]$, is the electrolyte. Ag^+ ions deposit on the article as a shiny silver layer.

Copper plating: the anode is a copper plate and the electrolyte is copper sulphate solution with a few drops of dilute sulphuric acid added to prevent its hydrolysis.

$$\text{At anode:} \quad Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$$

$$\text{At cathode:} \quad Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

In this way the concentration of $CuSO_4$ remains almost constant and a thin, even layer of copper is deposited on the article.

16.7 What is corrosion? Explain the reactions involved in corrosion of iron.

Corrosion is a natural process due to which a metal suffers gradual decay when it reacts chemically with its environment, and its surface becomes coated with compounds such as oxides, sulphides and carbonates. A common example is the rusting of iron, in which iron reacts with oxygen and water to form hydrated iron(III) oxide, which is rust. Rust is porous, so air and moisture keep reaching the fresh metal and the whole piece of iron slowly corrodes.

Rusting needs both oxygen and water, and it takes place through electrochemical reactions. Iron loses electrons and forms Fe(II) ions, oxygen gains these electrons in the presence of water and forms hydroxide ions, and then iron(II) hydroxide is oxidised further to iron(III) hydroxide, which is rust.

$$\text{At anode:} \quad 2Fe(s) \rightarrow 2Fe^{2+}(aq) + 4e^{-}$$

$$\text{At cathode:} \quad O_{2(g)} + 2H_2O(l) + 4e^{-} \rightarrow 4OH^{-}(aq)$$

$$\text{Then:} \quad Fe^{2+}(aq) + 2OH^{-}(aq) \rightarrow Fe(OH)_{2(s)}$$

$$4Fe(OH)_{2(s)} + O_{2(g)} + 2H_2O(l) \rightarrow 4Fe(OH)_{3(s)} \quad \text{(rust)}$$

Corrosion can be prevented by coating the metal surface with paint, grease, zinc (galvanisation), tin or another metal by electroplating, or by sacrificial protection in which a more reactive metal such as zinc or magnesium is allowed to corrode in place of iron.

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